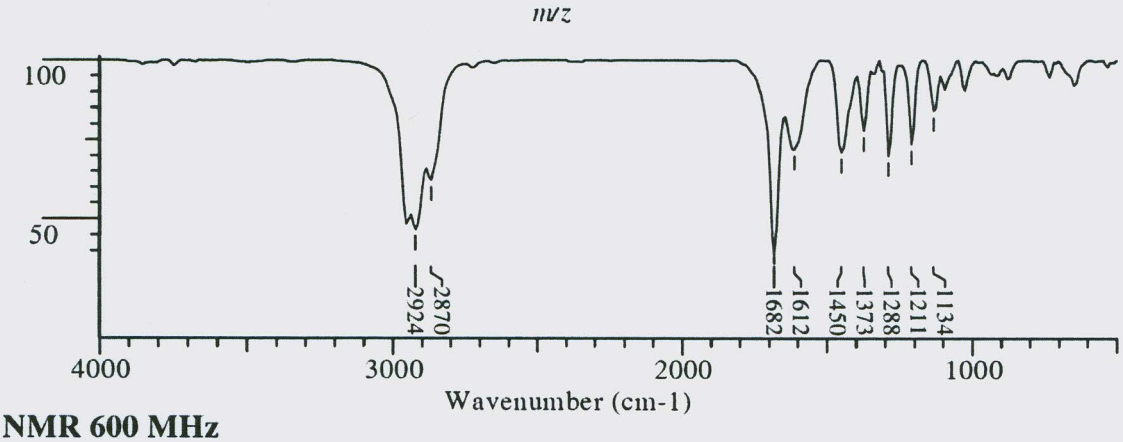
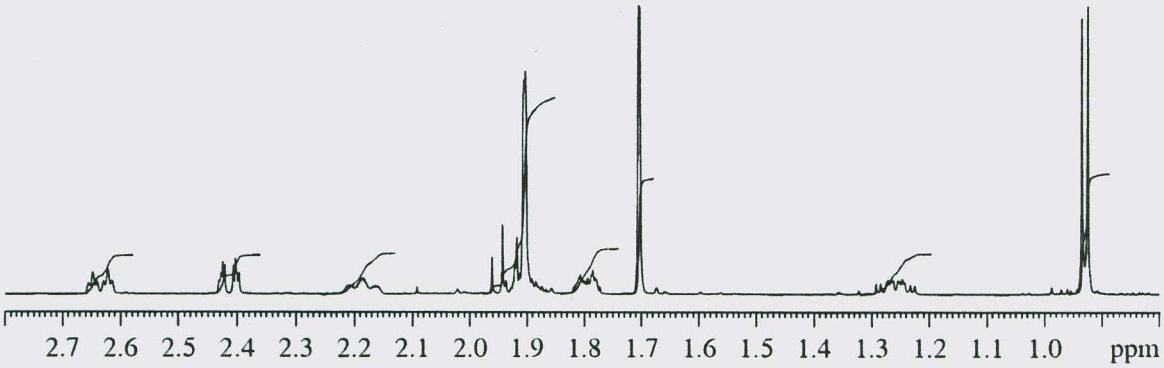
Organic Cumulative Exam
Feb 14, 2014

An unknown compound has the molecular formula of $C_{10}H_{16}O$. Its IR, proton, carbon (also DEPT), COSY, HMQC, and HMBC NMR spectra are reproduced below. Deduce the structure. Partial points are possible only for key fragments, **NOT** simple substituents. (Note: you need to judge by yourself whether a signal is reliable or not.)

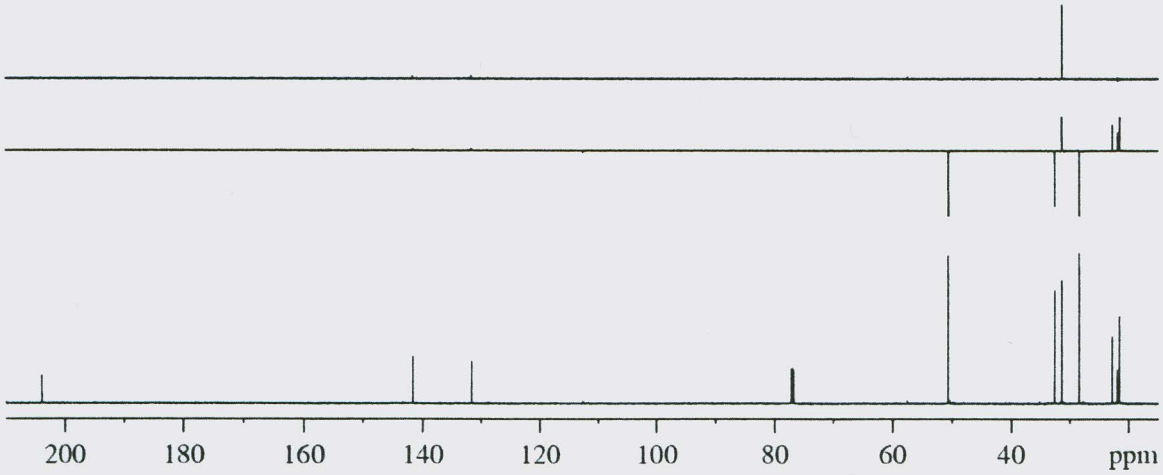
IR

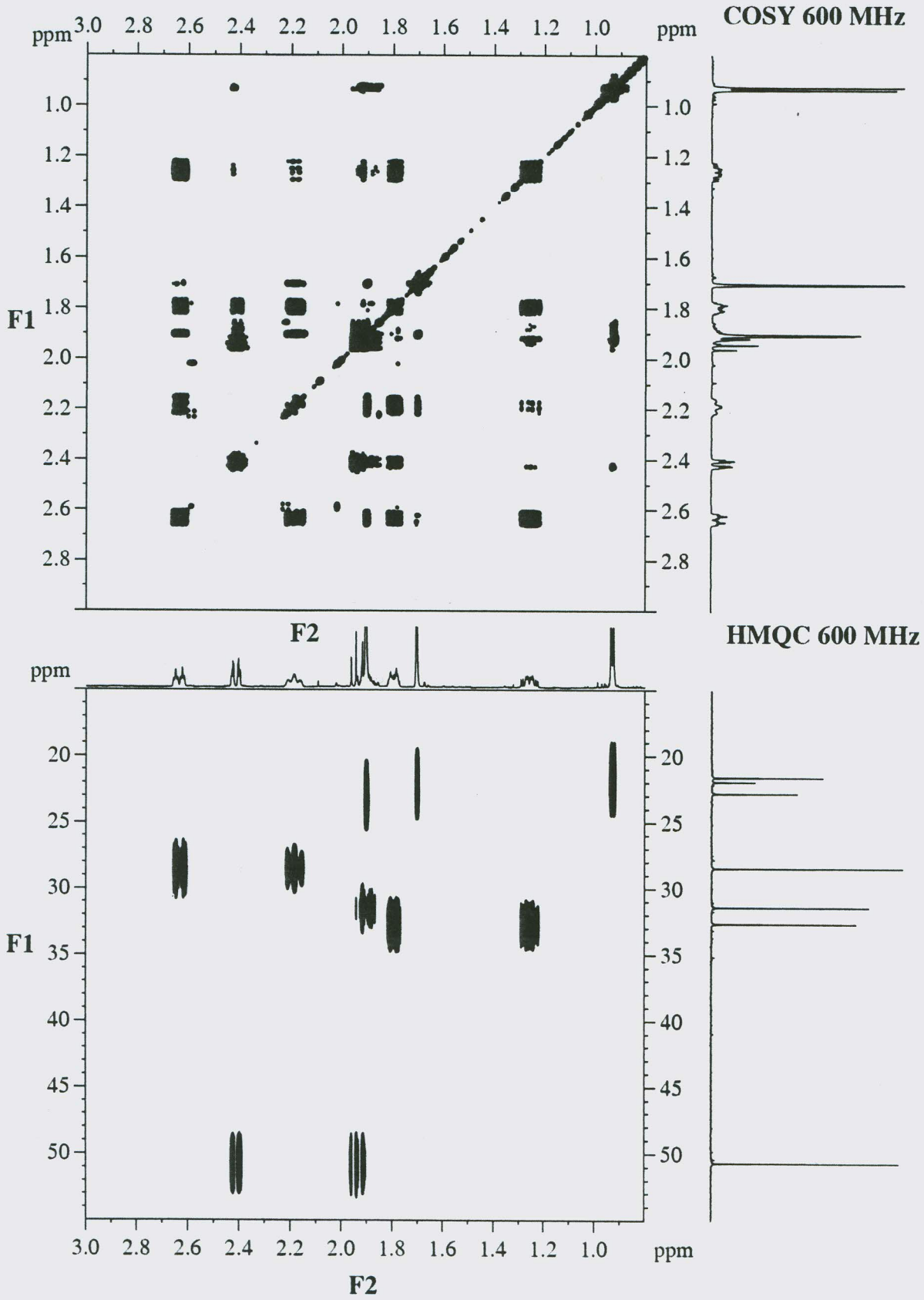


^{1}H NMR 600 MHz

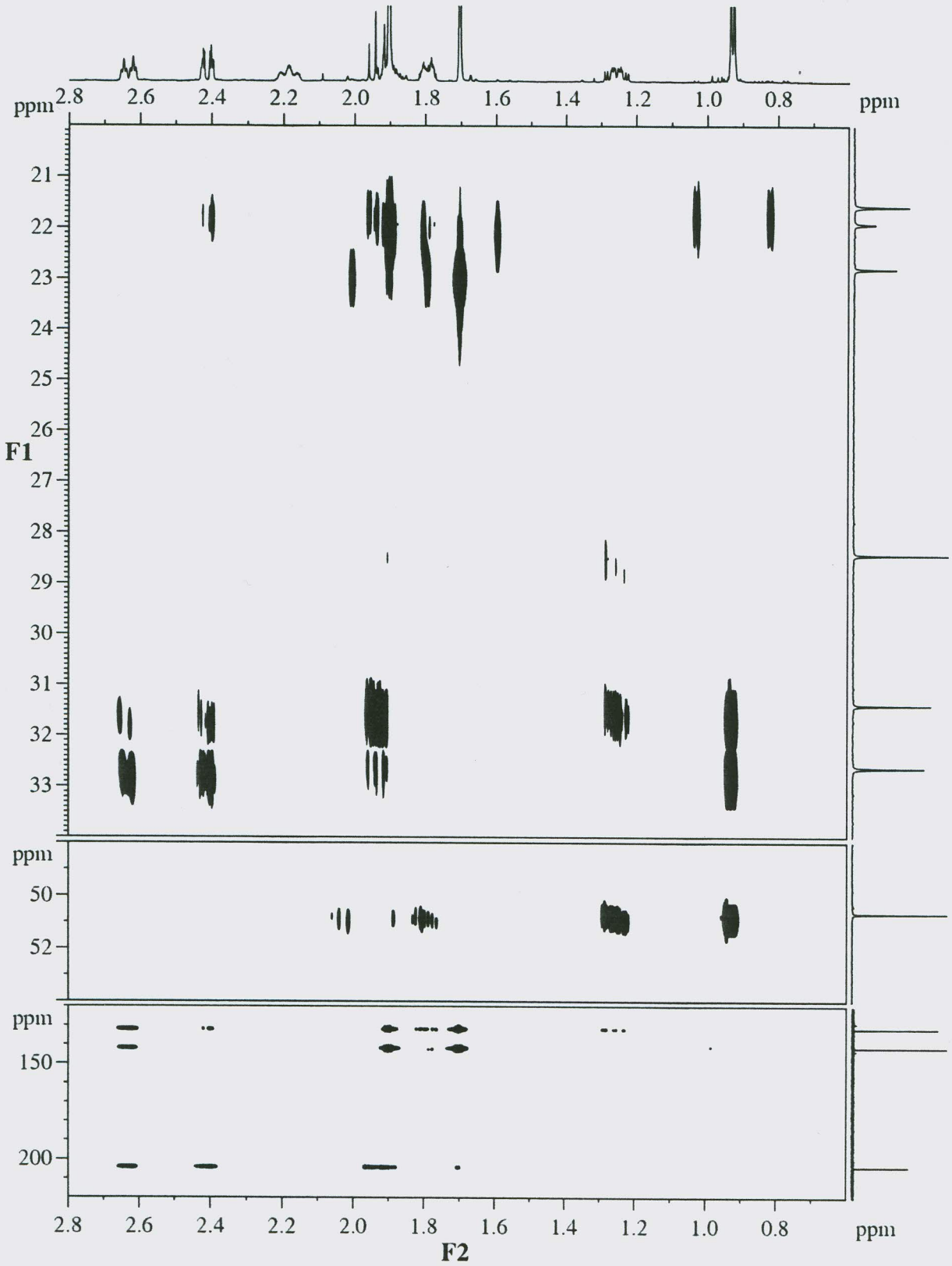


^{13}C/DEPT NMR 150.9 MHz




COSY 600 MHz
ppm
3.0 2.8 2.6 2.4 2.2 2.0 1.8 1.6 1.4 1.2 1.0
F1
F2
HMQC 600 MHz
20 25 30 35 40 45 50

HMBC 600 MHz


ppm
2.8
2.6
2.4
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
ppm
21
22
23
24
25
26
F1
27
28
29
30
31
32
33
ppm
50
52
ppm
150
200
2.8
2.6
2.4
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
ppm
F2

Spectral Reference Tables

Approximate Chemical Shifts

Type of proton	^1H δ (ppm)
1° alkyl, RCH_3	0.8-1.0
2° alkyl, RCH_2R	1.2-1.4
3° alkyl, R_3CH	1.4-1.7
allylic, C=C–CH_3	1.6-1.9
benzylic, $ArCH_3$	2.2-2.5
alkyl chloride, RCH_2Cl	3.6-3.8
alkyl bromide, RCH_2Br	3.4-3.6
alkyl iodide, RCH_2I	3.1-3.3
ether, $ROCH_2R$	3.3-3.9
alcohol, RCH_2OH	3.3-4.0
ketone, RC(=O)CH_3	2.1-2.6
aldehyde, RC(=O)H	9.5-9.6
vinylic, $R_2C=CH_2$	4.6-5.0
vinylic, $R_2C=CRH$	5.2-5.7
aromatic, ArH	6.0-9.5
acetylenic, RC≡CH	2.5-3.1
alcohol hydroxyl, ROH	0.5-6.0*
carboxylic acid, RCO_2H	10-13*
phenolic, ArOH	4.5-7.7*
amino, RNH_2	1.0-5.0*
amines, R_2NCH_3	2.2-2.6

Type of carbon	^{13}C δ (ppm)
1° alkyl, RCH_3	0-40
2° alkyl, RCH_2R	10-50
3° alkyl, R_3CH	15-50
alkyl halide or amine (sp^3) C–X (X = Cl, Br, N)	10-65
alcohol or ether (sp^3) C–O	50-90
alkenyl, =C	100-150
aryl	110-150
acetylenic, ≡C	70-90
carbonyl carbons:	
carboxylic acids, esters	160-185
amides	150-180
aldehydes, ketones	182-215

* chemical shift varies in different solvents and with temperature and concentration

Approximate Vibrational Frequencies

Functional Group Vibration	Frequency range (cm^{-1})	Intensity*
alkyl (sp^3) C–H stretching	2850-2960	m - s
alkenyl (sp^2) C–H stretching	3020-3100	m - s
Aromatic C–H out of plane bend		
monosubstituted benzene	690-710 and 730-770	
ortho disubstituted benzene	735-770	
meta disubstituted benzene	690-710 and 810-850	
para disubstituted benzene	810-840	
alkenyl C=C stretching	1640-1680	v
alkynyl ≡C–H stretching	~3300	s
alkynyl C≡C stretching	2100-2260	v
aromatic C=C stretching	1500 and 1600	s
aromatic (sp^2) C–H stretching	~3030	v
alcohols, phenols O–H stretching (dilute solutions)	3590-3650	s, sharp
alcohols, phenols O–H stretching (hydrogen bonded)	3400-3650	s, broad
carboxylic acids O–H stretching (hydrogen bonded)	2500-3000	s, very broad
amines N–H stretching	3300-3500	m
nitriles C≡N stretching	2220-2260	m
carbonyl C=O stretching:	1630-1780	s
aldehydes	1725 (sat'd); 1675-1700 (conj.)	s
ketones	1715 (sat'd); 1675-90 (conj.)	s
esters	1735 (sat'd); 1720 (conj.)	s
carboxylic acids	1760 (monomer); 1710 (dimer)	s
amides	1650-1690	s

* s=strong, m=medium, v=variable